MATLAB Starter Guide

Project 1 | Stage 1 Questions 6-7 | No previous MATLAB experience needed

Follow these steps to run the supplied file, make two plots and save your results. Use MATLAB R2020a or later. No extra toolbox or Simulink is needed.

1. Open MATLAB

Open MATLAB from the Windows Start menu or the Mac Applications folder. Use the university installation. If MATLAB is unavailable, ask your teacher or university IT for access.

2. Find the supplied file

Download the Project 1 folder. If it is a ZIP file, extract it first. Keep the MATLAB subfolder with the project files.

```
Project 1 folder
    MATLAB/
        Project01_01_Input.m
        Plots/    (created when the program runs)
```

3. Open the program

In MATLAB, click **Open** on the Home tab. Select `Project01_01_Input.m` from the MATLAB subfolder. The code opens in the **Editor**.

4. Choose the working folder

In the Files / Current Folder panel, browse to that same MATLAB subfolder. You should see the .m file in the list. Panel names and positions may vary by version.

Schematic panel layout; the screen may look different in your version.

5. Run the supplied program once

Click **Run** in the Editor. If MATLAB asks about the folder, choose **Change Folder**. Wait until the **>>** prompt returns in the Command Window.

No plot window? The supplied file uses `Visible = off`. Open **MATLAB/Plots/Project01_Input.png** to see its saved plot. The next page makes a visible two-panel plot.

Make the two plots

Work in your own copy of the supplied script

6. Save your own copy

Choose **Save As** in the Editor. Save in the same MATLAB subfolder as `Project01_Stage1_StudentID.m`. Replace StudentID with your ID. Keep the .m extension and use no spaces in the name.

7. Replace only the plotting and output section

In your copy, find the line beginning `f = figure(`. Select from that line to the end of the file. Replace that selection with the code below. Keep all earlier lines: they create `folder, t, T, r` and `rd`.

```
f = figure('Color','w','Position',[100 100 900 650]);

subplot(2,1,1);
plot(t,1000*r,'LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Floor displacement (mm)');
title('Floor input');
xlim([0 0.7]);

subplot(2,1,2);
plot(t,1000*rd,'LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Floor velocity (mm/s)');
xlim([0 0.7]);

exportgraphics(f,fullfile(folder,'Stage1_Input.png'), ...
    'Resolution',200);
writematrix([t r rd], ...
    fullfile(folder,'Stage1_Signals.csv'));
```

`subplot(2,1,1)` selects the top panel; `subplot(2,1,2)` selects the bottom panel. The supplied r and rd are in m and m/s. Multiplying by 1000 changes the plotted units to mm and mm/s.

8. Save and run your copy

Click **Save**, then **Run**. A figure opens with displacement above and velocity below. Open **MATLAB/Plots/Stage1_Input.png** to check the saved image. The file is replaced each time you run this copy.

Edit code in the Editor. Do not paste the whole script into the Command Window. Copy only the code, without page headings or line numbers.

Read values and times

An array holds many values; an index selects one value

The script already contains three arrays: **t** is time, **r** is floor displacement and **rd** is floor velocity. Values at the same index belong to the same instant.

9. Find a value and its position

For example, `[rPeak,iR] = max(r)` finds the largest displacement and its index iR. The matching time is `t(iR)`. Use `min` to find a minimum.

10. Add the numerical checks

Paste this block at the end of your modified script, below the code on page 2. Save, then click **Run**. Read the table and endpoint message in the **Command Window**.

```
[rPeak,iR] = max(r);
[vMax,iMax] = max(rd);
[vMin,iMin] = min(rd);

Quantity = {'Peak displacement'; ...
    'Maximum velocity'; 'Minimum velocity'};
Value = 1000*[rPeak; vMax; vMin];
Time_s = [t(iR); t(iMax); t(iMin)];
results = table(Quantity,Value,Time_s);
disp(results);
writetable(results,fullfile(folder,'Stage1_Peaks.csv'));

[~,iEnd] = min(abs(t-T));
fprintf('At t = %.3f s: r = %.6g mm, rd = %.6g mm/s\n', ...
    t(iEnd),1000*r(iEnd),1000*rd(iEnd));
```

Reading the table: the displacement row is in mm; the two velocity rows are in mm/s. All times are in seconds. **Stage1_Peaks.csv** is saved in the Plots folder. Open it in a spreadsheet or copy the results into your report.

11. Check the bump end

The last two commands select the time sample closest to T, the bump duration, and print both signals there. A tiny value such as `1e-12` means 0.000000000001; this can be numerical round-off.

12. Complete your own checks

Use the bump formula in the project to check the extrema and their times analytically. Explain the comparison in your report. This guide covers MATLAB operations; the input, states, model order and relative-travel questions still need your own answers.

In the report: use mm and mm/s to 1 decimal place, and short times to 3 decimal places in seconds. Keep full precision in calculations. A sampled peak time can differ slightly from the exact time.

Save and submit

Check the files before closing MATLAB

13. Check your outputs

In the MATLAB subfolder, open **Plots**. You should now have these files from your modified script:

File	What it contains
Stage1_Input.png	Your two-panel figure
Stage1_Peaks.csv	Three extrema and their times
Stage1_Signals.csv	Raw signals: time (s), r (m), rd (m/s)

14. Prepare the Stage 1 Questions 6-7 submission

Put the two-panel image, your results table and your analytical checks in the report. Include your answers to the model questions. Submit your modified **.m file** with the report. The CSV files help you prepare the report; this guide adds no extra submission requirement.

If something goes wrong

Problem	What to do
The file cannot be found	Check the Files / Current Folder panel. Open the MATLAB subfolder containing your saved .m file, then click Run.
No figure appears	Check that you are running your own copy. Use the figure line on page 2; do not keep Visible set to off.
Unrecognized variable t, r, rd or folder	Keep the lines above f = figure(...). Run the whole saved script, not just selected lines.
A red error message appears	Click the line number in the error. Check brackets, commas and straight quotation marks. Copy the relevant code block again if needed.
Only one panel, or old results	Check the active Editor filename. Save your changes and run that file again. Check the file in Plots.
exportgraphics is not recognized	The supplied program needs MATLAB R2020a or later. Use a university computer with a supported version.

Official help

MathWorks: Create Scripts | subplot | exportgraphics

Based on the current Project 1 Stage 1 Questions 6-7 and Project01_01_Input.m.